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MSBA Ops Multi-Agent Dispatch Report

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Dispatch Planning Report

Time-Critical Medicine Shipments | Next 48 Hours

1. Executive Summary

- Planning horizon: Day0 + Day1 (next 48 hours) for shipments from NJ distribution centers.
- Focus corridors: **C1_I95_NJ_BOS** and **C2_NJ_PHL**.
- Moderate weather risk (score 2) forecasted on Day1 with heavy rain and high wind at key waypoints.
- Recommended 25% travel time buffer applied on Day1 to mitigate weather impact without SLA violations.
- Resource allocation maintained: 2 drivers, 1 standard truck, 1 temperature-controlled truck per corridor.
- Zero SLA violations expected with current plan; no escalation required.

2. Key Performance Indicators (KPIs) by Corridor

C1_I95_NJ_BOS (Next 48h)

Total Shipments	Valid Units	Excluded Shipments	Exclusion Rate	Cold Chain Units	Cold Chain %	Tier 1 Units	Tier 1 %	Drivers Required	Standard Trucks	Temp-Controlled Trucks
18	16	2	11.1%	9	56.3%	12	75.0%	2	1	1

C2_NJ_PHL (Next 48h)

Total Shipments	Valid Units	Excluded Shipments	Exclusion Rate	Cold Chain Units	Cold Chain %	Tier 1 Units	Tier 1 %	Drivers Required	Standard Trucks	Temp-Controlled Trucks
15	14	1	6.7%	7	50.0%	11	78.6%	2	1	1

KPI Trends vs Historical Data

- Shipment volume:** Stable valid units with no absolute change.
- Cold chain units:** Slight decrease (-4%) compared to history.
- Tier 1 shipments:** Decreased by 15.4%, Tier 2 increased by 75%.
- Exclusion rates:** Stable at ~6-11% for next 48h vs ~2% historically.
- Resource needs:** No change in drivers or trucks required.

3. Data Quality & Anomalies

- DQ-01: 5 shipments missing unique_item_id excluded from dispatch calculations.
- DQ-02 to DQ-04: No invalid IDs, name mismatches, or duplicate UIDs detected.
- Excluded shipments primarily involve *Remdesivir* and *Insulin Lispro* with missing UIDs.
- Recommendation: Investigate flagged items to reduce future exclusions and improve data integrity.

Sample Excluded Shipments (Missing UID)

- C2_NJ_PHL: Insulin Lispro (item 10022)
- C1_I95_NJ_BOS: Remdesivir 100mg (item 10021) - 2 shipments
- C1_I95_NJ_BOS: Remdesivir 200mg (item 20021)

4. Weather Risk Summary

Corridor	Day0 Risk Score	Day1 Risk Score	48h Risk Score	Key Weather Risks (Day1)	Travel Buffer %	Escalation Required
C1_I95_NJ_BOS	0	2	2	Heavy Rain, High Wind	25%	No
C2_NJ_PHL	0	2	2	Heavy Rain, High Wind	25%	No

Key waypoints with weather risk on Day1 include Newark NJ, New Brunswick NJ, Trenton NJ, Bronx NY, and Philadelphia PA.

5. Dispatch Plan & Resource Allocation

- Maintain current allocation per corridor for Day0 and Day1:
 - 2 drivers
 - 1 standard truck
 - 1 temperature-controlled truck
- Apply 25% travel time buffer on Day1 to mitigate moderate weather risks.
- Prioritize Tier 1 cold-chain and room temperature shipments to ensure SLA compliance.
- Continue excluding shipments missing unique_item_id per DQ-01 and investigate flagged exclusions.
- No escalation required currently; standby resources recommended for potential weather deterioration.

6. Monitoring & Contingency Triggers

What to Monitor

- Real-time weather updates on Day1, focusing on heavy rain and high wind at critical waypoints.
- Shipment exclusions related to missing UIDs and flagged items (Remdesivir, Insulin Lispro).
- Tier 1 shipment volumes and cold-chain load to maintain priority handling.
- Resource utilization and availability of drivers and trucks.
- SLA compliance metrics for both corridors.

Contingency Triggers

- Weather risk score escalation to 3 (severe weather) requiring increased travel buffer and escalation.
- Unexpected resource shortages or unmet shipment units.
- Increase in shipment exclusions due to data quality issues.
- Detection of SLA violations in Tier 1 or Tier 2 shipments.

7. Expected KPI Impacts

- Zero SLA violations sustained with full Tier 1 shipment compliance.
- Stable exclusion rates with ongoing investigation of missing UID issues.
- Shipment volume and cold-chain load handled within allocated resources.
- No penalty points incurred from unmet demand or SLA breaches.
- Effective mitigation of weather-related delays via 25% travel buffer.

8. Audit Trail

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